

Bias Detection and Mitigation in Large Language Models

Authors: Dr. Priya Sharma, Prof. David Thompson, Dr. Aisha Patel

Journal: AI Ethics, Vol. 5, Issue 3

ABSTRACT:

We investigate gender, racial, and socioeconomic biases in popular large language models and propose novel debiasing techniques. Our comprehensive evaluation covers 8 models and 12 bias categories using both intrinsic and extrinsic evaluation metrics.

1. INTRODUCTION

Large language models have shown remarkable capabilities but often perpetuate harmful biases present in training data.

2. BIAS EVALUATION FRAMEWORK

2.1 Intrinsic Bias Measures

- Word Embedding Association Test (WEAT)
- Sentence Encoder Association Test (SEAT)
- Bias score calculation using cosine similarity

2.2 Extrinsic Bias Measures

- Downstream task performance across demographic groups
- Sentiment analysis bias in job descriptions
- Name-based discrimination in resume screening

2.3 Models Evaluated

GPT-3, GPT-4, BERT, RoBERTa, T5, PaLM, LaMDA, ChatGPT

3. DEBIASING TECHNIQUES

3.1 Data-level Interventions

- Balanced dataset curation
- Counterfactual data augmentation
- Bias-aware sampling strategies

3.2 Model-level Interventions

- Adversarial debiasing during training
- Post-processing bias correction
- Fairness-constrained optimization

4. RESULTS

Bias reduction achieved:

- Gender bias: 65% reduction
- Racial bias: 58% reduction
- Age bias: 72% reduction

Trade-offs: 3-5% decrease in downstream task performance